AI-Powered Student Learning Analytics and Performance Monitoring System

RESEARCH PROPOSAL

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Abstract

Educational institutions collect information about attendance, assessment results, assignment completion, and students' learning activities, but these data are not always combined into timely and understandable insights. This proposed final-year project will design and evaluate an AI-powered Student Learning Analytics and Performance Monitoring System that presents student progress through dashboards and estimates academic risk from available historical indicators. The study will investigate whether a prototype that combines descriptive analytics with a supervised machine-learning model can help identify students who may need additional academic support. A design-and-development approach will be used. Subject to institutional permission and data availability, the study will use a convenience or purposive sample of consenting students and/or anonymized academic records. Data fields may include attendance, assessment marks, assignment completion, and previous academic performance. The prototype will provide visual summaries, configurable performance indicators, and risk categories; model performance will be evaluated using suitable classification metrics such as precision, recall, F1-score, and accuracy, with particular attention to false negatives. A short usability questionnaire and task-based feedback may be used to assess clarity and usefulness. Privacy, consent, access control, and data minimization will guide the study. The expected outcome is a working prototype and an evaluation of its technical performance and perceived usefulness. Predictions will be treated as decision-support signals, not as definitive judgments about students, and the project's conclusions will be limited by dataset size, quality, and representativeness.

Introduction

Student performance is influenced by multiple factors, including attendance, assessment outcomes, assignment completion, and engagement with learning activities. When these indicators are reviewed separately or only after final examinations, educators may find it difficult to identify changing performance patterns early. Learning analytics applies the collection and analysis of data about learners and their contexts to understand and improve learning. Artificial intelligence (AI) and machine-learning techniques may support this process by detecting patterns in historical data and estimating which students may be at risk of low performance.

This project proposes a web-based system that combines student performance dashboards with a carefully evaluated predictive component. The intention is not to replace teachers' professional judgment, but to provide interpretable information that can support timely review and academic intervention. The proposal follows the sections in the supplied research-proposal template, including the problem statement, purpose, questions and hypothesis, definitions, theoretical framework, literature review, methodology, protection of human rights, and significance of the study.

Statement of the Problem

Academic information is often distributed across attendance registers, spreadsheets, learning platforms, and assessment records. This can make it difficult to view a student's progress over time, compare indicators consistently, or identify patterns that may warrant early support. Simple averages can summarize past results but may not reveal combinations of indicators associated with future academic difficulty.

The proposed study addresses the practical problem of bringing available academic indicators into one accessible monitoring interface and evaluating whether a machine-learning model can provide useful risk estimates from those indicators. The project will not assume that AI automatically improves educational outcomes. Instead, it will assess technical model performance and the perceived usefulness of the prototype, while acknowledging that data quality, sample size, bias, and local educational context may limit generalization.

Purpose of the Study

The purpose of this study is to design, develop, and evaluate an AI-powered student learning analytics system that visualizes academic progress and tests whether selected academic indicators can help classify students into predefined performance-risk categories.

The unit of analysis will be a student record for a defined academic period, where permission and suitable data are available. The system will also be assessed as a software prototype through functional testing and user feedback. Its practical importance is that teachers or academic advisers may receive a clearer summary of performance indicators and may review students who appear to need support earlier than they otherwise would.

Research Question/Hypothesis

The study will address the following research questions:

1. How can attendance, assessment results, assignment completion, and prior performance be combined into a dashboard that supports student performance monitoring?

2. How accurately can selected machine-learning methods classify students into predefined academic-risk categories using available historical indicators?

3. How do intended users rate the usability, clarity, and perceived usefulness of the proposed system?

Research hypothesis: A supervised machine-learning model trained on suitable historical academic indicators will classify the study dataset’s predefined performance-risk categories better than a simple majority-class baseline. This hypothesis will be evaluated using held-out data and appropriate classification metrics. It is not a claim that the system will necessarily improve grades or learning outcomes.

Definition of Terms

1. Learning analytics — the measurement, collection, analysis, and reporting of data about learners and their contexts to understand and optimize learning and the environments in which it occurs (Siemens, 2013).

2. Student performance monitoring — the organized review of indicators such as attendance, assessment marks, assignment completion, and progress over time.

3. Academic-risk category — a predefined label, derived from an institution-approved rule or available outcome data, indicating that a student may require further academic review. It is not a diagnosis or a final judgment about a student.

4. Machine learning — computational methods that learn patterns from data and use those patterns to make predictions or classifications.

5. Dashboard — a visual interface that summarizes key indicators and trends to help users inspect information.

6. Early-warning indicator — a measurable signal that may prompt a teacher or adviser to review whether a student needs support.

Theoretical Framework

The project will be guided by the learning analytics cycle: collecting relevant learning data, analyzing it, presenting interpretable findings, and using those findings to inform human decisions. This framework fits the project because the proposed system must do more than generate a prediction; it must present information in a form that educators can interpret and act on responsibly.

The design will also draw on the educational data-mining perspective described by Romero and Ventura (2020), in which educational data are analyzed to discover patterns that may support understanding and improvement of learning. The framework predicts that combining transparent summaries with carefully evaluated predictive indicators can make academic information easier to inspect. Whether users find the system useful, and whether the model performs adequately, will be tested rather than assumed.

Literature Review

Learning analytics and educational decision support

Siemens (2013) describes learning analytics as an emerging field concerned with making sense of learner data to understand and optimize learning. This provides a conceptual basis for a system that combines data, analysis, and presentation for educational decision support. For the proposed project, the important implication is that analytics should be linked to an understandable educational purpose rather than collecting data without a clear use.

Educational data mining and prediction

Romero and Ventura (2020) review the development of educational data mining and learning analytics, including the use of data-driven techniques to examine educational processes. Their review supports investigating machine-learning methods for educational datasets, while also highlighting the diversity of data, tasks, and evaluation settings in the field. The proposed project will therefore compare model performance against a simple baseline and report the dataset and evaluation limitations clearly.

Predicting academic performance

Research in educational data mining has examined relationships between student characteristics, learning activity, and academic outcomes. Such work suggests that historical assessment and engagement indicators can be useful inputs, but the strength of any prediction depends on the quality and representativeness of the data and on how the target outcome is defined. This project will use only indicators approved for the study and will avoid treating correlations as proof that an indicator causes a particular result.

Identified practical gap

For this final-year project, the practical gap is the need for a small, integrated prototype that presents understandable performance trends alongside a transparently evaluated prediction component. The project's contribution will be a locally testable system design and an evaluation using the data that can legitimately be obtained. The literature review should be expanded with additional institution-specific and recent empirical studies after the supervisor confirms the final scope and available dataset.

Methodology

A design-and-development methodology will be used to build a working software prototype and evaluate its main functions. The study will combine quantitative model evaluation with descriptive user feedback. The exact dataset, sample size, and risk-label definition will be finalized after supervisor approval and confirmation of institutional permissions.

Research Design

The project will use a prototype-development and evaluation design. The planned stages are: (1) confirm requirements and permissions; (2) identify approved data fields and define the target risk category; (3) clean and prepare the data; (4) develop the dashboard and prediction workflow; (5) test functionality and model performance; and (6) collect user feedback, analyze results, and document limitations.

Candidate inputs include attendance percentage, assignment completion, assessment marks, and previous academic results, subject to availability and approval. Candidate models may include logistic regression and decision tree or random forest classifiers. A simple baseline will be included for comparison. The data will be separated into training and test sets, or evaluated with cross-validation if the dataset is small enough to justify that approach. Any preprocessing will be fitted using training data only to reduce data leakage.

Sampling

The target population is students enrolled in the participating course, program, or institution for which permission and appropriate data are available. A feasible initial sample is approximately 50–100 student records, but the final number will depend on access, data completeness, and supervisor guidance. Convenience or purposive sampling may be used because participation and access are constrained by institutional approval. The sample's academic level and other relevant characteristics will be described only where ethically approved and necessary. If an adequate dataset cannot be obtained, the project will use a clearly documented, permitted anonymized dataset or revise the predictive scope with the supervisor; it will not fabricate student records and present them as real observations.

Instrumentation

Instrument 1: Academic data extraction sheet

A structured data dictionary and extraction sheet will define each permitted field, its format, source, and handling rules. It will be used to check completeness, consistency, and missing values. Only fields necessary for the research questions will be collected.

Instrument 2: Software prototype and test checklist

The prototype will include a role-appropriate dashboard, performance summaries, trend visualizations, and a prediction output where sufficient data exist. A test checklist will record whether core functions work as specified, including data validation, access restrictions, calculations, and display of results. Functional testing will establish whether the software behaves as intended; it will not by itself prove educational effectiveness.

Instrument 3: Model evaluation measures

The prediction component will be evaluated using a confusion matrix and, as appropriate, accuracy, precision, recall, and F1-score. Recall for the at-risk category will receive particular attention because missed students may be important in an early-warning setting. Results will be compared with a simple baseline and reported with the test procedure and class distribution. If the dataset is too small or lacks reliable labels, predictive claims will be limited or the model component will be reframed as exploratory.

Instrument 4: User feedback questionnaire

A short questionnaire may ask intended users to rate ease of use, clarity of visualizations, perceived usefulness, and confidence in understanding the displayed information. Responses will be summarized descriptively. If the questionnaire is newly developed for this project, its results will be treated as exploratory rather than as evidence of a previously validated scale.

Data Collection and Analysis Procedures

After receiving the required institutional permission and informed consent where applicable, approved academic records will be obtained and de-identified before analysis. Direct identifiers such as student names and roll numbers will not be used as model features. Data will be checked for missing, inconsistent, or duplicate values. The target outcome and any thresholds used to define risk will be documented before model evaluation to avoid choosing a definition merely because it produces favorable results.

Descriptive analysis will summarize the available indicators and their distributions. The prediction model will be evaluated on held-out data or through a suitable cross-validation procedure, depending on dataset size. Accuracy, precision, recall, F1-score, and the confusion matrix will be reported where applicable; results will be compared with a baseline. User questionnaire responses will be summarized using counts, percentages, and average ratings where appropriate. The prototype will also undergo functional testing. No statistical significance claim will be made unless the design and sample support it.

Protection of Human Rights

Participation in questionnaires or user testing will be voluntary. Participants will be told the purpose of the study, what data are collected, how the data will be used, and that they may withdraw without penalty. Required institutional approval and consent will be obtained before accessing identifiable or restricted academic records. Data will be minimized, de-identified where possible, stored securely, and made available only to authorized project personnel. Findings will be reported in aggregate and will not publicly identify individual students.

The system’s predictions will be presented as prompts for human review, not as final decisions about students. Users should be able to understand the indicators behind a risk flag, and the project will acknowledge that errors and bias are possible. Data will not be used to punish students or make high-stakes decisions within this research project. A summary of the completed study can be provided to interested participants or the institution, subject to confidentiality requirements.

Discussion / Significance of the Study

The proposed system may benefit students and educators by making academic progress easier to review and by providing a structured way to identify records that may warrant further attention. For teachers or advisers, a combined dashboard could reduce the effort required to compare separate indicators. For students, earlier human review may create an opportunity for timely guidance, although this project will not claim improved outcomes without a suitable longitudinal evaluation.

The project’s expected academic contribution is a documented prototype and an evidence-based account of its usability and predictive performance on the available dataset. Likely limitations include a small or non-representative sample, missing records, inconsistent data definitions, class imbalance, and limited time for evaluation. Internal validity will be supported through clear target definitions, data-preparation documentation, and held-out evaluation; external validity will remain limited unless the system is tested across different courses or institutions. Future work could investigate larger datasets, model calibration, fairness across relevant groups where ethically and statistically appropriate, and whether interventions informed by the system improve outcomes.

Proposed Work Plan

The following is a tentative 25-day plan, subject to supervisor feedback, institutional permissions, and data availability:

Days 1–3: Confirm scope, requirements, ethics/privacy requirements, and dataset access.

Days 4–6: Review literature, define variables and target outcome, and prepare the data dictionary.

Days 7–10: Clean and explore permitted data; establish a baseline and initial model evaluation.

Days 11–16: Build dashboard, data-processing workflow, and prediction interface.

Days 17–19: Conduct functional testing and evaluate model performance.

Days 20–21: Collect user feedback if approval and participants are available; summarize findings.

Days 22–25: Improve the prototype, write the report, prepare demonstration materials, and incorporate supervisor feedback.

References

Baker, R. S., & Inventado, P. S. (2014). Educational data mining and learning analytics. In J. A. Larusson & B. White (Eds.), Learning analytics: From research to practice (pp. 61–75). Springer. https://doi.org/10.1007/978-1-4614-3305-7_4

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Siemens, G. (2013). Learning analytics: The emergence of a discipline. American Behavioral Scientist, 57(10), 1380–1400. https://doi.org/10.1177/0002764213498851

Note: The references above provide an initial scholarly foundation. Before submission, verify the required citation style with the department and expand the literature review with additional recent studies relevant to the approved dataset and system scope.